t3-w02 — condition C0

Plane Shapes (2-D Shapes) — Plane shapes: types, properties and perimeter

> The curriculum objective alone. No lesson plan, no note.

INPUT LAYERS

Input layer	What it carries	Channel	In this condition
Scope	what is to be taught, and to whom	prompt	**present**
Prior knowledge	what the learner already knew coming in	prompt	absent
Materials	what the lesson was represented with	prompt	absent
Position	where this objective sat in the sequence	prompt	absent
Method	the shape of the teaching, as event labels	prompt	absent
Enacted content	what was actually explained and worked through	prompt	absent
Set work	what the learners were asked to do in class	prompt	absent
Assessment	the questions the lesson was checked with	prompt	absent
Figures	the drawings the lesson used	tool	absent

WHAT THE MODEL IS SENT

Contract `ComposeWriteObjectiveSection`, declaring: `artefact_spec, lesson_context, unit_context`.

Lesson context — shared by every section

`subject`

Mathematics

`grade_level`

JSS1

`topic`

Plane Shapes (2-D Shapes)

`subtopic`

Plane shapes: types, properties and perimeter

`term`

Third Term

`week`

2

`lesson_objectives`

• identify the types of plane shapes and state their properties, similarities and differences
• identify the parts of a circle
• find the perimeter of a regular polygon, square, rectangle, trapezium, parallelogram and circle

Objective 1 — identify the types of plane shapes and state their properties, similarities and differences

`objective_text`

identify the types of plane shapes and state their properties, similarities and differences

Withheld from this payload: `step_title`, `step_content`, `instructional_type`, `estimated_minutes`, `current_step`, `instructional_events`, `student_tasks`, `activities`, `assessment_points`, `source_evidence`, `source_figures`, `objective_learner_state` — carried by the plan or the unit, not sent under this condition.

Objective 2 — identify the parts of a circle

`objective_text`

identify the parts of a circle

Withheld from this payload: `step_title`, `step_content`, `instructional_type`, `estimated_minutes`, `current_step`, `instructional_events`, `student_tasks`, `activities`, `assessment_points`, `source_evidence`, `source_figures`, `objective_learner_state` — carried by the plan or the unit, not sent under this condition.

Objective 3 — find the perimeter of a regular polygon, square, rectangle, trapezium, parallelogram and circle

`objective_text`

find the perimeter of a regular polygon, square, rectangle, trapezium, parallelogram and circle

Withheld from this payload: `step_title`, `step_content`, `instructional_type`, `estimated_minutes`, `current_step`, `instructional_events`, `student_tasks`, `activities`, `assessment_points`, `source_evidence`, `source_figures`, `objective_learner_state` — carried by the plan or the unit, not sent under this condition.

WHAT THE MODEL PRODUCED

Objective 1

Exploring Plane Shapes: Types, Properties and How They Compare

EXPLANATION

A *plane shape* is a flat figure that lies in a single flat surface. The most common types you will meet are:

1. **Triangle** – three sides and three angles. Types of triangles (equilateral, isosceles, scalene) differ by side lengths and angle sizes.
2. **Quadrilateral** – four sides and four angles. Important quadrilaterals include:
   - **Square** – four equal sides, four right angles.
   - **Rectangle** – opposite sides equal, four right angles.
   - **Parallelogram** – opposite sides equal and parallel, opposite angles equal.
   - **Rhombus** – four equal sides, opposite angles equal (angles need not be right).
   - **Trapezium (trapezoid)** – at least one pair of parallel sides.
3. **Pentagon, Hexagon, …** – regular polygons have all sides and all interior angles equal.
4. **Circle** – a set of points equidistant from a centre; it has no sides or angles.

Key properties to remember
- *Sides*: count how many straight edges a shape has.
- *Angles*: sum of interior angles depends on the number of sides (e.g., quadrilaterals always add to 360°).
- *Parallel sides*: shapes like rectangles, squares, parallelograms, and trapeziums have pairs of parallel sides.
- *Equal sides*: squares and rhombuses have four equal sides; rectangles have opposite sides equal.

Similarities
- All quadrilaterals share four sides and four angles, and their interior angles total 360°.
- Squares, rectangles, and rhombuses are all *parallelograms* because opposite sides are parallel.

Differences
- Only a square has both all sides equal *and* all right angles.
- A rectangle has right angles but only opposite sides equal.
- A rhombus has all sides equal but its angles are not necessarily right.
- A trapezium has only one pair of parallel sides, unlike the other quadrilaterals.
- A circle is completely different: it has no straight sides or angles.

Understanding these properties helps you identify a shape quickly and compare it with others.

WORKED_EXAMPLE

Worked Example: Comparing a Square and a Rectangle

Suppose you have two shapes:
- Shape A is a *square* with side length 5 cm.
- Shape B is a *rectangle* with length 8 cm and width 5 cm.

Property	Square (A)	Rectangle (B)
Number of sides	4	4
Side lengths	all 5 cm	two sides 8 cm, two sides 5 cm
Angles	4 right angles	4 right angles
Parallel sides	yes (all opposite sides)	yes (both pairs)
Equal sides?	yes	no (only opposite sides equal)

What does this tell you?
- Both shapes are quadrilaterals and have four right angles, so they share those properties.
- The square is a special type of rectangle because it meets all the rectangle's conditions *and* has all sides equal.
- If you only know a shape has four right angles and opposite sides equal, you cannot be sure whether it is a rectangle or a square unless you also check the side lengths.

This comparison shows how shapes can be similar in some ways and different in others.

1. 1. List three properties that all quadrilaterals share.
answer: They have four sides, four angles, and the interior angles add up to 360°.

2. 2. Which of the following shapes is *not* a quadrilateral? a) rhombus b) triangle c) trapezium d) rectangle
answer: b) triangle

3. 3. A shape has four equal sides and four right angles. What is the name of this shape?
answer: Square

4. 4. True or False: Every rectangle is also a parallelogram.
answer: True

5. 5. Identify one key difference between a rhombus and a square.
answer: A square has four right angles, while a rhombus does not necessarily have right angles.

6. 6. Which shape has no straight sides or angles? a) circle b) pentagon c) trapezium d) parallelogram
answer: a) circle

Objective 2

Parts of a Circle

EXPLANATION
A circle is made up of several important parts:

- **Centre**: The fixed point in the middle of the circle. Every point on the circle is the same distance from the centre.
- **Radius**: A line segment from the centre to any point on the circle. All radii have the same length.
- **Diameter**: A line that passes through the centre and whose endpoints lie on the circle. It is twice the length of a radius.
- **Circumference**: The distance around the circle, often called the perimeter of a circle.
- **Chord**: A line segment whose endpoints both lie on the circle but does not necessarily pass through the centre.
- **Arc**: A part of the circumference between two points on the circle.
- **Sector**: The region bounded by two radii and the arc between them (like a slice of pizza).
- **Segment**: The region bounded by a chord and the arc it subtends.

Remember: every radius is equal, the diameter is the longest chord, and the circumference can be found using the formula \(C = 2\pi r\) or \(C = \pi d\).

WORKED_EXAMPLE
Worked Example

Find the diameter, radius and circumference of a circle whose radius is 5 cm.

1. **Radius (r)** is given as 5 cm.
2. **Diameter (d)** = 2 × r = 2 × 5 cm = 10 cm.
3. **Circumference (C)** = 2πr = 2 × π × 5 cm ≈ 31.4 cm (using π ≈ 3.14).

So, the circle has a radius of 5 cm, a diameter of 10 cm, and a circumference of about 31.4 cm.

1. 1. In a circle, the distance from the centre to any point on the circle is called the _____.
answer: radius

2. 2. A line that goes through the centre and touches the circle at two points is the _____.
answer: diameter

3. 3. If the radius of a circle is 7 cm, what is its circumference? (Use π = 3.14)
answer: 43.96 cm

4. 4. The longest possible chord of a circle is the _____.
answer: diameter

5. 5. A segment of the circle's edge between two points is called an _____.
answer: arc

Objective 3

Finding Perimeters of Common Shapes

> EXPLANATION
> The perimeter of a shape is the total distance around its edge. For the shapes you need to work with, the formulas are:
>
> - **Regular polygon (n sides, each of length s)**: \(P = n \times s\)
> - **Square (side a)**: \(P = 4a\)
> - **Rectangle (length l, width w)**: \(P = 2(l + w)\)
> - **Trapezium (sides a, b, c, d)**: \(P = a + b + c + d\) (add all four side lengths)
> - **Parallelogram (base b, side s)**: \(P = 2(b + s)\)
> - **Circle (radius r)**: \(P = 2\pi r\) (or \(P = \pi d\) if you know the diameter)
>
> Remember to keep the units consistent and write your answer with the appropriate unit (e.g., cm, m).

> WORKED_EXAMPLE
> **Worked Example – Rectangle**
>
> A rectangle has a length of 8 cm and a width of 5 cm. What is its perimeter?
>
> Solution:
> \[P = 2(l + w) = 2(8\text{ cm} + 5\text{ cm}) = 2(13\text{ cm}) = 26\text{ cm}\]
> So the perimeter of the rectangle is **26 cm**.

1. A square has a side length of 7 cm. Find its perimeter.
answer: 28 cm

2. A regular hexagon (6-sided polygon) has each side measuring 4 cm. What is its perimeter?
answer: 24 cm

3. A rectangle measures 12 cm by 9 cm. Determine its perimeter.
answer: 42 cm

4. The sides of a trapezium are 5 cm, 7 cm, 8 cm and 6 cm. Find its perimeter.
answer: 26 cm

5. A parallelogram has a base of 10 cm and a side length of 6 cm. What is its perimeter?
answer: 32 cm

6. A circle has a radius of 3 cm. Calculate its perimeter (use \(\pi \approx 3.14\)).
answer: 18.84 cm

LAYER BY LAYER: WHAT WAS GROUNDED AND WHAT WAS NOT

Output layer	Made	Written from	Verdict
Title the reader-facing framing of the section	3	Scope	**grounded**
Explanation the principle, rule or relationship to review	3	nothing in the input	**unsourced**
Worked example a problem with its solution worked through	3	nothing in the input	**unsourced**
Figure a drawing the learner reads from	0	nothing in the input	not produced
Practice what the learner does	17	nothing in the input	**unsourced**
Answer what the learner checks against	17	nothing in the input	**unsourced**